

Microprocessor and Computer Architecture

BCA — Semester 2 — — Answer Sheet

Subject Code:

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Group B

1. What is 8085 microprocessor? Explain the functional units of 8085 microprocessor in brief.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

2. Draw the block diagram of Hardwired control unit and explain their functional units in brief.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. State an algorithm for the restoring division method of fixed point binary division. Show the step-by-step division process using restoring division algorithm when 448(01110 0000) is divided by 17(10001). Use 5-bit register to represent the numbers.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

4. What is the difference between SAP-1 and SAP-2?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. What is instruction cycle? If the clock frequency is 5 MHz, how much time is required to execute an instruction MVI A, 08H?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. What is Microoperations? Explain the function of each following microoperations $AR \leftarrow PC$, $DR \leftarrow M[AR]$, $PC \leftarrow PC + 1$, $AR \leftarrow DR(0 - 10)$, $CA(2 - 5) \leftarrow DR(11 - 14)$

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. What is Instruction Format of basic computer architecture? A computer uses a memory unit with 256K words of 32 bits each. Draw the instruction word format and indicate the number of bits in each part.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. Write a 8085 assembly language for division of two 8-bit numbers by repeated subtraction method.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. Explain the use of mapping table.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. What is the difference between isolated I/O and memory-mapped I/O? What are the advantages and disadvantages of each?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. What is pipelining? Explain the role of pipelining in computing.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

12. Write short notes on: a. Multiplexed Address b. Memory Hierarchy

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.